Statistical Physics

Lecture Notes & Formula Sheets

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0.1 The Microcanonical Ensemble (MCE)

0.1.1 Basic Definition

The Microcanonical Ensemble describes an isolated system with a fixed total energy E, fixed volume V, and fixed number of particles N. The fundamental assumption of statistical mechanics states that for an isolated system in equilibrium, all accessible microstates are equally probable.

If $\Omega(E)$ represents the number of microstates with energy E, the probability $P^{(n)}$ of finding the system in a specific microstate n is given by:

$$P^{(n)} = \begin{cases} \frac{1}{\Omega(E)} & \text{if } E_n = E \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

The connection to thermodynamics is established through the Boltzmann entropy formula:

$$S(E) = k_B \ln \Omega(E) \tag{2}$$

where k_B is the Boltzmann constant.

0.1.2 Interaction Between Two Systems

Consider a combined system consisting of two subsystems, System 1 and System 2, which are in thermal contact but isolated from the rest of the universe. The total energy is conserved:

$$E_{\text{tot}} = E_1 + E_2 = \text{constant} \tag{3}$$

The total number of microstates for the combined system, $\Omega(E_{\text{tot}})$, is the sum over all possible partitions of energy between the two subsystems:

$$\Omega(E_{\text{tot}}) = \sum_{\{E_1\}} \Omega_1(E_1)\Omega_2(E_{\text{tot}} - E_1) \tag{4}$$

Since the number of states typically grows exponentially with the size of the system, the sum is dominated by the term where the product $\Omega_1\Omega_2$ is maximized. To find the equilibrium condition, we maximize the entropy (or equivalently, the number of states) with respect to E_1:

$$\frac{d}{dE_1}\left[\Omega_1(E_1)\Omega_2(E_{\text{tot}} - E_1)\right] = 0 \tag{5}$$

Taking the logarithm (since $S = k_B \ln \Omega$) simplifies the derivative:

$$\frac{\partial}{\partial E_1}\left[S_1(E_1) + S_2(E_{\text{tot}} - E_1)\right] = 0 \tag{6}$$

Using the chain rule, this yields the condition for thermal equilibrium:

$$\frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2} \implies \frac{1}{T_1} = \frac{1}{T_2} \tag{7}$$

Thus, temperature is rigorously defined as:

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N} \tag{8}$$

0.2 The Canonical Ensemble (CE)

0.2.1 System in Contact with a Heat Reservoir

We now consider a system S in thermal contact with a much larger heat reservoir R. The combined system is isolated with total fixed energy $E_{\text{tot}} = E_S + E_R$.

The probability $P^{(n)}$ that the system S is in a specific microstate n with energy E_n is proportional to the number of microstates available to the reservoir, given that the reservoir has energy $E_{\text{tot}} - E_n$:

$$P^{(n)} \propto \Omega_R(E_{\text{tot}} - E_n) \tag{9}$$

Using the definition of entropy, we can rewrite this as:

$$P^{(n)} \propto \exp\left(\frac{S_R(E_{\text{tot}} - E_n)}{k_B}\right) \tag{10}$$

0.2.2 Derivation of the Boltzmann Factor

Since the reservoir is very large compared to the system ($E_{\text{tot}} \gg E_n$), we can perform a Taylor expansion of the reservoir's entropy S_R around the total energy E_{tot}:

$$S_R(E_{\text{tot}} - E_n) \approx S_R(E_{\text{tot}}) - E_n\left(\frac{\partial S_R}{\partial E}\right)_{E=E_{\text{tot}}} + \dots \tag{11}$$

Identifying the derivative as the inverse temperature $\beta \equiv \frac{1}{k_B T}$:

$$S_R(E_{\text{tot}} - E_n) \approx S_R(E_{\text{tot}}) - \frac{E_n}{T} \tag{12}$$

Substituting this expansion back into the probability expression:

$$P^{(n)} \propto \exp\left(\frac{S_R(E_{\text{tot}})}{k_B}\right) \exp\left(-\frac{E_n}{k_B T}\right) \tag{13}$$

The first term is a constant (independent of the specific state n). Therefore, the probability is:

$$P^{(n)} = \frac{1}{Z} e^{-\beta E_n} \tag{14}$$

where Z is the normalization constant, known as the **Partition Function**:

$$Z = \sum_n e^{-\beta E_n} \tag{15}$$

0.3 Gibbs Entropy of the Canonical Ensemble from Microcanonical Principles

We can derive the Canonical Ensemble (CE) probability distribution starting from the statistical definition of the Microcanonical Ensemble (MCE). This approach uses the concept of an "ensemble of ensembles" and the Variational Principle.

0.3.1 Statistical Weight and Gibbs Entropy

Consider a supersystem consisting of a large number $\mathcal{W}$ of identical systems. Let n_i be the number of systems occupying the microstate i. The probability of finding a system in state i is given by the frequency ratio:

$$P^{(i)} = \frac{n_i}{\mathcal{W}} \tag{16}$$

The number of ways $\Omega_{\mathcal{W}}$ (statistical weight) to arrange these systems among the available states is given by the multinomial coefficient:

$$\Omega_{\mathcal{W}} = \frac{\mathcal{W}!}{\prod_i n_i!} \tag{17}$$

Using the Boltzmann entropy definition $S_{\text{total}} = k_B \ln \Omega_{\mathcal{W}}$ and applying Stirling's approximation ($\ln N! \approx N \ln N - N$) for large $\mathcal{W}$:

$$\begin{aligned}
\ln \Omega_{\mathcal{W}} &= \ln \mathcal{W}! - \sum_i \ln n_i! && (18)\\
&\approx (\mathcal{W} \ln \mathcal{W} - \mathcal{W}) - \sum_i (n_i \ln n_i - n_i) && (19)\\
&= \mathcal{W} \ln \mathcal{W} - \sum_i n_i \ln n_i \quad (\text{since } \sum n_i = \mathcal{W}) && (20)\\
&= -\sum_i n_i \ln\left(\frac{n_i}{\mathcal{W}}\right) = -\mathcal{W} \sum_i P^{(i)} \ln P^{(i)} && (21)
\end{aligned}$$

The entropy per system is then the famous **Gibbs Entropy** formula:

$$S = \frac{S_{\text{total}}}{\mathcal{W}} = -k_B \sum_i P^{(i)} \ln P^{(i)} \tag{22}$$

Note on Consistency: If we assume the Microcanonical condition where all accessible states $\Omega(E)$ are equally probable (i.e., $P^{(i)} = 1/\Omega(E)$), the Gibbs entropy reduces to the Boltzmann entropy:

$$S = -k_B \sum_{i=1}^{\Omega} \frac{1}{\Omega} \ln\left(\frac{1}{\Omega}\right) = -k_B \left(\Omega \cdot \frac{1}{\Omega}\right)(-\ln \Omega) = k_B \ln \Omega(E) \tag{23}$$

0.4 Example: The Two-State System

Consider a system of N non-interacting particles, where each particle can be in one of two states:

- Ground state: energy 0
- Excited state: energy ϵ

Let N_g be the number of particles in the ground state and N_e be the number of particles in the excited state ($N = N_g + N_e$). The total energy of the system is $E = N_e\epsilon$.

0.4.1 Microcanonical Approach

The number of ways to choose N_e excited particles out of N is given by the binomial coefficient:

$$\Omega(E) = \frac{N!}{N_e!(N - N_e)!} \tag{24}$$

The entropy is $S = k_B \ln \Omega$. Using Stirling's approximation ($\ln N! \approx N \ln N - N$), we can derive the temperature relationship:

$$\frac{1}{T} = \frac{\partial S}{\partial E} \implies \frac{N_e}{N} = \frac{1}{e^{\beta\epsilon} + 1} \tag{25}$$

0.4.2 Canonical Approach

The partition function for a single particle is:

$$Z_1 = 1 + e^{-\beta\epsilon} \tag{26}$$

For N distinguishable particles, the total partition function is $Z = (Z_1)^N$. The average energy can be calculated using $\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$, yielding the same result as the Microcanonical approach.

0.5 Energy Fluctuations and Ensemble Equivalence

0.5.1 Calculating Fluctuations

In the Canonical Ensemble, the energy is not fixed but fluctuates around a mean value $\langle E \rangle$. The variance of the energy is defined as:

$$\Delta E^2 = \langle (E - \langle E \rangle)^2 \rangle = \langle E^2 \rangle - \langle E \rangle^2 \tag{27}$$

We can derive this directly from the partition function. Recall that $\langle E \rangle = \frac{1}{Z} \sum E_n e^{-\beta E_n}$. Differentiating $\langle E \rangle$ with respect to β:

$$\frac{\partial \langle E \rangle}{\partial \beta} = \frac{\partial}{\partial \beta} \left(\frac{\sum E_n e^{-\beta E_n}}{Z} \right) = \langle E \rangle^2 - \langle E^2 \rangle \tag{28}$$

Therefore, the variance is related to the specific heat capacity C_V:

$$\langle (\Delta E)^2 \rangle = -\frac{\partial \langle E \rangle}{\partial \beta} = k_B T^2 C_V \tag{29}$$

0.5.2 Thermodynamic Limit

We define the relative fluctuation as $\frac{\sqrt{\langle (\Delta E)^2 \rangle}}{\langle E \rangle}$. Since energy and heat capacity are extensive quantities (proportional to N), we have $\langle E \rangle \sim N$ and $C_V \sim N$.

$$\frac{\Delta E_{rms}}{E} = \frac{\sqrt{k_B T^2 C_V}}{E} \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}} \tag{30}$$

As $N \to \infty$ (the thermodynamic limit), the relative fluctuations vanish ($\Delta E / E \to 0$). The energy distribution becomes a sharp Dirac delta function centered at the mean energy.

$$Z \approx \Omega(E_m) e^{-\beta E_m} \tag{31}$$

This mathematically proves that the Canonical and Microcanonical ensembles yield identical physical predictions for macroscopic systems.

0.6 Variational Principle: Ensemble Derivation

The different statistical ensembles can be derived mathematically by maximising the Gibbs entropy subject to specific constraints. This is known as the principle of maximum entropy.

0.6.1 Microcanonical Ensemble (MCE)

For an isolated system, the only constraint is that the sum of probabilities must be normalized. We do not constrain the average energy because the energy is fixed at E for all accessible states (i.e., $P_n = 0$ if $E_n \neq E$).

Constraint:

$$\sum_n P_n = 1 \tag{32}$$

To maximize entropy under this constraint, we define the Lagrangian function $\mathcal{L}$ with a Lagrange multiplier α:

$$\mathcal{L} = -\sum_n P_n \ln P_n - \alpha \left(\sum_n P_n - 1 \right) \tag{33}$$

Note: We drop the constant k_B for the maximization step as it does not affect the position of the maximum. We set the derivative with respect to P_n to zero:

$$\frac{\partial \mathcal{L}}{\partial P_n} = -(1 + \ln P_n) - \alpha = 0 \tag{34}$$

Solving for P_n:

$$\ln P_n = -(1+\alpha) \implies P_n = e^{-(1+\alpha)} = \text{constant} \tag{35}$$

Since $\sum P_n = 1$ and P_n is constant for all Ω accessible states, we recover the fundamental assumption of the Microcanonical Ensemble:

$$P_n = \frac{1}{\Omega} \tag{36}$$

0.6.2 Canonical Ensemble (CE)

For a system in thermal equilibrium with a heat bath, the energy is allowed to fluctuate, but the *average* energy $\langle E \rangle$ is fixed by the temperature of the bath.

Constraints:

1. Normalization: $\sum_n P_n = 1$
2. Fixed Average Energy: $\sum_n P_n E_n = \langle E \rangle$

We introduce two Lagrange multipliers: α (for normalization) and β (for energy). The Lagrangian is:

$$\mathcal{L} = -\sum_n P_n \ln P_n - \alpha \left(\sum_n P_n - 1 \right) - \beta \left(\sum_n P_n E_n - \langle E \rangle \right) \tag{37}$$

Differentiating with respect to P_n and setting to zero:

$$\frac{\partial \mathcal{L}}{\partial P_n} = -(1 + \ln P_n) - \alpha - \beta E_n = 0 \tag{38}$$

Solving for $\ln P_n$:

$$\ln P_n = -(1+\alpha) - \beta E_n \tag{39}$$

Exponentiating both sides gives the Canonical distribution:

$$P_n = e^{-(1+\alpha)} e^{-\beta E_n} \tag{40}$$

We identify the normalization constant $e^{-(1+\alpha)}$ as $1/Z$. Thus:

$$P_n = \frac{1}{Z} e^{-\beta E_n} \tag{41}$$

where $Z = \sum_n e^{-\beta E_n}$ and β is identified physically as $1/k_B T$.

Concept	**Formula**
Microcanonical Ensemble *(Fixed E, V, N)*	
Microstates (Weight)	$\Omega(E) =$ Number of accessible states
Boltzmann Entropy	$S = k_B \ln \Omega(E)$
Definition of Temperature	$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N}$
Canonical Ensemble *(Fixed T, V, N)*	
Partition Function	$Z = \sum_n e^{-\beta E_n} \quad (\beta \equiv \frac{1}{k_B T})$
Canonical Probability	$P^{(n)} = \frac{1}{Z} e^{-\beta E_n}$
Average Energy	$\langle E \rangle = -\frac{\partial \ln Z}{\partial \beta}$
Fluctuations & Thermodynamic Limit	
Energy Fluctuation	$\langle (\Delta E)^2 \rangle = k_B T^2 C_V$
Relative Fluctuation	$\frac{\Delta E_{\text{rms}}}{\langle E \rangle} \sim \frac{1}{\sqrt{N}}$

Table 1: Summary of Key Formulas: Microcanonical vs. Canonical

Chapter 1

Statistical Mechanics of Ideal Gases

To calculate the partition function of an ideal gas, we first need to know the number of available quantum states in a given energy or momentum range. This is described by the **density of states** $g(k)$.

1.1 Derivation of Density of States

1.1.1 Method A: Fixed Boundary Conditions (Standing Waves)

Consider a particle in a cubic box of side L. The wavefunction must vanish at the boundaries. The wavenumber k is quantized as:

$$k_i = \frac{\pi n_i}{L} \quad (n_i = 1, 2, 3, \dots) \tag{1.1}$$

The states form a grid in the positive octant of k-space. The density of states is derived by counting the number of points in a spherical shell of radius k and thickness dk:

$$g(k)dk = \underbrace{\frac{1}{8}}_{\text{Octant}} \times \underbrace{4\pi k^2 dk}_{\text{Shell Volume}} \times \underbrace{\frac{1}{(\pi/L)^3}}_{\text{Density of points}} \tag{1.2}$$

$$g(k)dk = \frac{Vk^2}{2\pi^2}dk \tag{1.3}$$

1.1.2 Method B: Periodic Boundary Conditions (Traveling Waves)

Alternatively, we can use periodic boundary conditions: $\psi(x + L, y, z) = \psi(x, y, z)$. The free particle wavefunction is a plane wave:

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{\sqrt{V}} e^{ik_x x} e^{ik_y y} e^{ik_z z} \tag{1.4}$$

Periodicity implies $e^{ik_x L} = 1$, so the quantization condition is:

$$k_x = \frac{2\pi n_x}{L} \quad (n_x = 0, \pm 1, \pm 2, \dots) \tag{1.5}$$

Here, the volume of a single k-state is $(2\pi/L)^3$. Since n_x can be negative, we consider the full sphere (not just 1/8):

$$g(k)dk = \frac{4\pi k^2 dk}{(2\pi/L)^3} = \frac{Vk^2}{2\pi^2}dk \tag{1.6}$$

Both methods yield the **same result**.

1.2 The Canonical Partition Function

1.2.1 Single-Particle Partition Function (Z_1)

The energy of a free particle is purely kinetic:

$$E_k = \frac{\hbar^2 k^2}{2m} \tag{1.7}$$

The single-particle partition function is the integral of the Boltzmann factor over all states:

$$Z_1 = \int_0^\infty e^{-\beta E_k} g(k) dk = \frac{V}{2\pi^2} \int_0^\infty e^{-\frac{\beta \hbar^2 k^2}{2m}} k^2 dk \tag{1.8}$$

Using the standard Gaussian integral $\int_0^\infty x^2 e^{-ax^2} dx = \frac{\sqrt{\pi}}{4a^{3/2}}$:

$$Z_1 = \frac{V}{\lambda_{th}^3} \tag{1.9}$$

where we have defined the **Thermal de Broglie Wavelength**:

$$\lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T}} \tag{1.10}$$

This length scale represents the quantum "fuzziness" of the particle.

1.2.2 N-Particle Partition Function (Z_N)

For N **distinguishable** particles, $Z_N = (Z_1)^N$. However, atoms of the same gas are **indistinguishable**. We must divide by $N!$ to avoid overcounting permutations:

$$Z_N = \frac{1}{N!}(Z_1)^N = \frac{V^N}{N! \lambda_{th}^{3N}} \tag{1.11}$$

1.3 Indistinguishability and The Gibbs Paradox

The factor of $1/N!$ is crucial for resolving the Gibbs Paradox. Consider the mixing of gases :

1.3.1 Case 1: Distinguishable Gases (Different species)

If we mix Gas A and Gas B (both starting at V, T into total volume $2V$), the entropy change is:

$$\Delta S = S_f - S_i = 2N k_B \ln 2 \tag{1.12}$$

This is the **Entropy of Mixing**. It is an irreversible process.

1.3.2 Case 2: Indistinguishable Gases (Identical species)

If we mix Gas A with more Gas A, there is no physical change. The entropy change must be zero.

- **Without** $1/N!$**:** We would incorrectly calculate $\Delta S = 2Nk_B \ln 2$.
- **With** $1/N!$**:** The partition function correctly accounts for indistinguishability, yielding:

$$\Delta S = 0 \tag{1.13}$$

This confirms that the mixing of identical gases is a reversible process (in fact, it's no process at all).

1.4 Internal Degrees of Freedom: Diatomic Gases

For a diatomic molecule, the partition function factorizes into independent degrees of freedom:

$$Z = Z_{\text{trans}} \times Z_{\text{vib}} \times Z_{\text{rot}} \tag{1.14}$$

1.4.1 Translational Contribution

$$Z_{\text{trans}} = \frac{V}{\lambda_{th}^3} \implies U = \frac{3}{2}Nk_BT \implies C_V = \frac{3}{2}Nk_B \tag{1.15}$$

1.4.2 Vibrational Contribution

Modeled as a Quantum Harmonic Oscillator:

$$Z_{\text{vib}} = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+1/2)} = \frac{e^{-\beta\hbar\omega/2}}{1 - e^{-\beta\hbar\omega}} \tag{1.16}$$

1.4.3 Rotational Contribution

Modeled as a Rigid Rotor with energy levels $E_J = \frac{\hbar^2}{2I}J(J+1)$:

$$Z_{\text{rot}} = \sum_{J=0}^{\infty}(2J+1)e^{-\frac{\beta\hbar^2}{2I}J(J+1)} \tag{1.17}$$

At high temperatures, the sum can be approximated by an integral, leading to $U_{rot} = Nk_BT$ (linear molecule has 2 axes of rotation).

1.4.4 Total Heat Capacity

Summing the contributions (at moderate temperatures where vibration is frozen out but rotation is active):

$$C_V = \underbrace{\frac{3}{2}R}_{\text{Trans}} + \underbrace{R}_{\text{Rot}} = \frac{5}{2}R \tag{1.18}$$

1.5 Summary of Ideal Gas Statistical Mechanics

Symbol	**Formula**
DOS (k-space)	$g(k)dk = \frac{Vk^2}{2\pi^2}dk$
Thermal λ_{th}	$\lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T}}$
Single Part. Z_1	$Z_1 = \frac{V}{\lambda_{th}^3}$
N-Part. Z_N	$Z_N = \frac{1}{N!}(Z_1)^N = \frac{1}{N!}\left(\frac{V}{\lambda_{th}^3}\right)^N$
Helmholtz Energy F	$F = -k_B T \ln Z_N$
Internal Energy U	$U = -\frac{\partial \ln Z}{\partial \beta}$
entropy S	$S = -\left(\frac{\partial F}{\partial T}\right)_V = \frac{U-F}{T}$

Chapter 2

The Grand Canonical Ensemble & Chemical Potential

The chemical potential μ characterizes how the energy of a system changes when we add particles to it.

2.1 Thermodynamic Definitions

Starting from the fundamental thermodynamic relation for internal energy $U(S, V, N)$:

$$dU = TdS - pdV + \mu dN \tag{2.1}$$

We can define μ as the partial derivative:

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} \tag{2.2}$$

Physically, μ represents the energy cost to add one more particle to the system while keeping entropy S and volume V fixed.

We can also define μ using other thermodynamic potentials via Legendre transformations:

- **Helmholtz Free Energy ($F = U - TS$):**

$$dF = -SdT - pdV + \mu dN \implies \mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} \tag{2.3}$$

- **Gibbs Free Energy ($G = F + pV$):**

$$dG = -SdT + Vdp + \mu dN \implies \mu = \left(\frac{\partial G}{\partial N}\right)_{T,p} \tag{2.4}$$

2.2 Conditions for Equilibrium

Consider two systems in contact.

1. **Thermal Equilibrium (Heat Exchange):** The entropy change for energy exchange dU is $dS = (\frac{1}{T_1} - \frac{1}{T_2})dU$. Maximizing entropy ($dS = 0$) implies $T_1 = T_2$.

2. **Diffusive Equilibrium (Particle Exchange):** The entropy change for particle exchange dN is:

$$dS = \left(\frac{\partial S_1}{\partial N_1}\right)_{U,V} dN_1 + \left(\frac{\partial S_2}{\partial N_2}\right)_{U,V} dN_2 \tag{2.5}$$

Using $\frac{\mu}{T} = -(\frac{\partial S}{\partial N})_{U,V}$ and conservation $dN_1 = -dN_2$:

$$dS = \left(\frac{\mu_2}{T_2} - \frac{\mu_1}{T_1}\right) dN_1 \geq 0 \tag{2.6}$$

If $T_1 = T_2$, equilibrium requires $\mu_1 = \mu_2$. Particles flow from high μ to low μ.

2.3 Chemical Potential of an Ideal Gas

For an ideal gas, the canonical partition function is $Z_N = \frac{1}{N!}(\frac{V}{\lambda_{th}^3})^N$. The Helmholtz free energy is:

$$F = -k_B T \ln Z_N \approx N k_B T[\ln(n\lambda_{th}^3) - 1] \tag{2.7}$$

where $n = N/V$ is the number density and λ_{th} is the thermal wavelength. Calculating μ:

$$\mu = \left(\frac{\partial F}{\partial N}\right)_{T,V} = k_B T \ln(n\lambda_{th}^3) \tag{2.8}$$

Since $G = F + pV$ and $pV = Nk_B T$:

$$G = Nk_B T[\ln(n\lambda_{th}^3) - 1] + Nk_B T = Nk_B T \ln(n\lambda_{th}^3) \tag{2.9}$$

Comparing the expressions for μ and G, we see that for an ideal gas:

$$\mu = \frac{G}{N} \tag{2.10}$$

2.4 Derivation of the Grand Canonical Ensemble

When a system can exchange both energy and particles with a reservoir (fixed T and μ), the number of particles N fluctuates.

2.4.1 Probability Distribution

Consider a system coupled to a large reservoir with total energy U_{tot} and particles N_{tot}. The probability of the system being in state i with energy E_i and particles N_i is proportional to the number of microstates of the reservoir:

$$P_i \propto \Omega_{res}(U_{tot} - E_i, N_{tot} - N_i) \tag{2.11}$$

Expanding the entropy $S = k_B \ln \Omega$ to first order:

$$S(U - E, N - N_i) \approx S(U, N) - E\frac{\partial S}{\partial U} - N_i\frac{\partial S}{\partial N} = S_{res} - \frac{E_i}{T} + \frac{\mu N_i}{T} \tag{2.12}$$

Thus, the probability distribution is:

$$P_i = \frac{1}{\mathcal{Z}} e^{\beta(\mu N_i - E_i)} \tag{2.13}$$

2.4.2 Grand Partition Function ($\mathcal{Z}$)

The normalization factor is the Grand Canonical Partition Function:

$$\mathcal{Z}(T, \mu, V) = \sum_i e^{-\beta(E_i - \mu N_i)} = \sum_{N=0}^{\infty} e^{\beta\mu N} Z_N(T, V) \tag{2.14}$$

2.4.3 Grand Potential (Φ_G)

Analogous to F in the canonical ensemble, we define the Grand Potential:

$$\Phi_G = -k_B T \ln \mathcal{Z} \tag{2.15}$$

Using the definitions of P_i and entropy $S = -k_B \sum P_i \ln P_i$, we can show:

$$\Phi_G = F - \mu N = U - TS - \mu N \tag{2.16}$$

2.5 Scaling Arguments and Gibbs-Duhem

2.5.1 Chemical Potential as Gibbs Free Energy

Since entropy S is an extensive variable, scaling the system size by λ scales all extensive variables (U, V, N) by λ:

$$\lambda S(U, V, N) = S(\lambda U, \lambda V, \lambda N) \tag{2.17}$$

Differentiating with respect to λ and setting $\lambda = 1$ (Euler's Theorem for homogeneous functions):

$$S = \frac{\partial S}{\partial U} U + \frac{\partial S}{\partial V} V + \frac{\partial S}{\partial N} N \tag{2.18}$$

Substituting thermodynamic derivatives $(\frac{1}{T}, \frac{p}{T}, -\frac{\mu}{T})$:

$$S = \frac{1}{T} U + \frac{p}{T} V - \frac{\mu}{T} N \implies U - TS + pV = \mu N \tag{2.19}$$

The LHS is exactly the Gibbs Free Energy G. Thus, we prove generally:

$$\mu = \frac{G}{N} \tag{2.20}$$

2.5.2 The Grand Potential Relation

Substitute $G = \mu N$ into the definition of the Grand Potential:

$$\Phi_G = F - \mu N = (G - pV) - \mu N = (\mu N - pV) - \mu N \tag{2.21}$$

$$\Phi_G = -pV \tag{2.22}$$

This provides a direct way to calculate the equation of state from the grand partition function: $p = \frac{k_B T}{V} \ln \mathcal{Z}$.

2.6 Special Case: Non-Conserved Particles

For systems where particle number is not conserved (e.g., photons in a cavity, phonons in a solid), particles are created or destroyed to minimize the free energy.

$$\left(\frac{\partial F}{\partial N}\right)_{T,V} = 0 \tag{2.23}$$

By definition of chemical potential, this implies:

$$\mu = 0 \tag{2.24}$$

This is why the Planck distribution for photons uses the Bose-Einstein distribution with $\mu = 0$.

2.7 Summary of Thermodynamic Potentials & Ensembles

Symbol	**Formula**
Chem. Potential μ	$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} = \left(\frac{\partial F}{\partial N}\right)_{T,V} = \left(\frac{\partial G}{\partial N}\right)_{T,p}$
Gibbs Energy G	$G = \mu N$
Grand Potential Φ_G	$\Phi_G = F - \mu N = -pV$
Grand Canonical Prob.	$P_i = \frac{1}{\mathcal{Z}} e^{-\beta(E_i - \mu N_i)}$
Canonical Part. Z	$Z = \sum_i e^{-\beta E_i} \implies F = -k_B T \ln Z$
Grand Part. $\mathcal{Z}$	$\mathcal{Z} = \sum_i e^{-\beta(E_i - \mu N_i)} \implies \Phi_G = -k_B T \ln \mathcal{Z}$

Chapter 3

The Photon Gas: Blackbody Radiation

Before deriving the statistics of blackbody radiation, we must establish the physical model of the electromagnetic field in a cavity. In Quantum Mechanics, the electromagnetic field is treated as a collection of photons, where the energy is quantized as $E = \hbar\omega$.

3.1 Conceptual Foundation: The Harmonic Oscillator Analogy

A fundamental concept in quantum electrodynamics is that the electromagnetic field can be mathematically treated as a collection of independent harmonic oscillators.

Problem: Why treat the EM field as harmonic oscillators?

To see this clearly, we examine the energy of the classical electromagnetic field using Vector Calculus and Fourier analysis.

We work in the **Coulomb Gauge** ($\nabla \cdot \mathbf{A} = 0$) with scalar potential $\phi = 0$. The fields are defined by the magnetic vector potential $\mathbf{A}(\mathbf{r}, t)$:

$$\mathbf{E} = -\frac{\partial \mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A} \tag{3.1}$$

3.1.1 Derivation of the Hamiltonian

Step 1: Expansion of the Field into Modes
Consider the field confined in a volume V. We expand $\mathbf{A}$ into a sum of spatial modes $\mathbf{u}_\mathbf{k}(\mathbf{r})$ and time-dependent amplitudes $q_\mathbf{k}(t)$:

$$\mathbf{A}(\mathbf{r}, t) = \sum_\mathbf{k} q_\mathbf{k}(t)\mathbf{u}_\mathbf{k}(\mathbf{r}) \tag{3.2}$$

We assume the spatial functions $\mathbf{u}_\mathbf{k}$ are orthonormal standing waves satisfying:

$$\int_V \mathbf{u}_\mathbf{k} \cdot \mathbf{u}_{\mathbf{k}'} \, dV = \delta_{\mathbf{k}\mathbf{k}'} \tag{3.3}$$

Step 2: Electric Field Energy (The "Kinetic" Term)
The total electric energy is the integral of the energy density:

$$U_E = \int_V \frac{\epsilon_0}{2}\mathbf{E}^2\, dV = \frac{\epsilon_0}{2}\int_V \left(-\sum_{\mathbf{k}} \dot{q}_{\mathbf{k}}\mathbf{u}_{\mathbf{k}}\right)^2 dV \tag{3.4}$$

Expanding the square involves a double sum $\sum_{\mathbf{k}}\sum_{\mathbf{k}'}$. However, due to the **orthogonality condition**, all cross terms ($\mathbf{k} \neq \mathbf{k}'$) vanish upon integration. We are left with:

$$U_E = \frac{\epsilon_0}{2}\sum_{\mathbf{k}} \dot{q}_{\mathbf{k}}^2 \tag{3.5}$$

This looks exactly like the kinetic energy term ($\frac{1}{2}mv^2$) of a mechanical system.

Step 3: Magnetic Field Energy (The "Potential" Term)
The magnetic energy is:

$$U_B = \int_V \frac{1}{2\mu_0}\mathbf{B}^2\, dV = \frac{1}{2\mu_0}\int_V (\nabla \times \mathbf{A}) \cdot (\nabla \times \mathbf{A})\, dV \tag{3.6}$$

Using the vector identity $\int(\nabla \times \mathbf{A})^2 dV = \int \mathbf{A} \cdot [-\nabla^2\mathbf{A}]dV$ (under Coulomb gauge and boundary conditions) and the Helmholtz equation $\nabla^2\mathbf{u}_{\mathbf{k}} = -k^2\mathbf{u}_{\mathbf{k}} = -(\omega_k^2/c^2)\mathbf{u}_{\mathbf{k}}$:

$$U_B = \frac{1}{2\mu_0}\int_V \mathbf{A} \cdot \sum_{\mathbf{k}'} q_{\mathbf{k}'}\left(\frac{\omega_{\mathbf{k}'}^2}{c^2}\mathbf{u}_{\mathbf{k}'}\right) dV \tag{3.7}$$

Using $c^2 = 1/(\epsilon_0\mu_0)$ and orthogonality again, the cross terms vanish:

$$U_B = \frac{\epsilon_0}{2}\sum_{\mathbf{k}} \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \tag{3.8}$$

This corresponds to the potential energy term ($\frac{1}{2}kx^2$) of a spring.

Conclusion:
The total Hamiltonian is the sum of independent harmonic oscillators:

$$H_{EM} = \sum_{\mathbf{k}}\left[\frac{\epsilon_0}{2}\dot{q}_{\mathbf{k}}^2 + \frac{\epsilon_0}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2\right] \tag{3.9}$$

By rescaling variables, this is identical to $H = \sum \hbar\omega(n + 1/2)$ in quantum mechanics.

3.2 Blackbody Radiation Statistics

Now we apply statistical mechanics to these photon modes.

- **Quantum Case:** $E = \hbar\omega$.
- **Density of States:** The number of modes in frequency range $d\omega$ (including 2 polarization states) is:

$$g(\omega)d\omega = \frac{V\omega^2}{\pi^2 c^3} d\omega \tag{3.10}$$

3.2.1 Partition Function

Since photons are bosons and the number of photons is not conserved (chemical potential $\mu = 0$), the partition function for a single mode with frequency ω is a geometric series:

$$Z_\omega = \sum_{n=0}^{\infty} e^{-\beta n\hbar\omega} = 1 + e^{-\beta\hbar\omega} + \cdots = \frac{1}{1 - e^{-\beta\hbar\omega}} \tag{3.11}$$

The log of the total partition function is the integral over all modes:

$$\ln Z = \int_0^\infty g(\omega) \ln Z_\omega \, d\omega = -\frac{V}{\pi^2 c^3} \int_0^\infty \omega^2 \ln(1 - e^{-\beta\hbar\omega}) \, d\omega \tag{3.12}$$

3.2.2 The Planck Distribution

The total internal energy U is derived from $U = -\frac{\partial \ln Z}{\partial \beta}$:

$$U = \int_0^\infty d\omega \, g(\omega) \langle E_\omega \rangle \tag{3.13}$$

where $\langle E_\omega \rangle$ is the average energy of a mode. We find:

$$U = \frac{V\hbar}{\pi^2 c^3} \int_0^\infty \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \, d\omega \tag{3.14}$$

The integrand defines the **spectral energy density** $u(\omega)$:

$$u(\omega)d\omega = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1} d\omega \tag{3.15}$$

Physical Interpretation: $u(\omega)d\omega$ represents the energy carried by photons with frequencies between ω and $\omega + d\omega$. This can also be written as the density of states times the average energy per mode:

$$u(\omega)d\omega = \underbrace{\left(\frac{\omega^2}{\pi^2 c^3}\right)}_{\text{Density of States } g(\omega)/V} \times \underbrace{(\hbar\omega)}_{\text{Photon Energy}} \times \underbrace{\frac{1}{e^{\beta\hbar\omega} - 1}}_{\text{Bose-Einstein Occ. } \langle N \rangle} d\omega \tag{3.16}$$

3.2.3 Total Energy and Stefan-Boltzmann Law

To find the total energy density $u = U/V$, we integrate over all frequencies. Let $x = \beta\hbar\omega$, then $d\omega = \frac{k_B T}{\hbar} dx$:

$$u = \frac{(k_B T)^4}{\pi^2 c^3 \hbar^3} \int_0^\infty \frac{x^3}{e^x - 1} dx \tag{3.17}$$

Using the standard integral $\int_0^\infty \frac{x^3}{e^x-1} dx = \frac{\pi^4}{15}$:

$$u(T) = \frac{\pi^2 k_B^4}{15\hbar^3 c^3} T^4 \tag{3.18}$$

The Stefan-Boltzmann Law: The energy flux (power per unit area) radiated by a blackbody is related to the energy density by geometry ($J = \frac{1}{4}cu$):

$$J = \sigma T^4 \tag{3.19}$$

where σ is the Stefan-Boltzmann constant.

3.3 Comparison: The Classical Limit (Ultraviolet Catastrophe)

In the classical limit (low frequency or high temperature), $\hbar\omega \ll k_B T$, so $e^{\beta\hbar\omega} \approx 1 + \beta\hbar\omega$. The expected energy of one mode becomes:

$$\langle E \rangle_{classical} \approx \frac{\hbar\omega}{\beta\hbar\omega} = k_B T \tag{3.20}$$

3.3.1 Derivation from Equipartition Theorem

Why is the average energy $k_B T$?
This result is a direct consequence of the classical **Equipartition Theorem**. Recall the Hamiltonian we derived for a single electromagnetic mode $\mathbf{k}$:

$$H_{\mathbf{k}} = \underbrace{\frac{\epsilon_0}{2}\dot{q}_{\mathbf{k}}^2}_{\text{Kinetic-like}} + \underbrace{\frac{\epsilon_0}{2}\omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2}_{\text{Potential-like}} \tag{3.21}$$

This Hamiltonian contains **two** independent quadratic degrees of freedom ($n = 2$). According to the Equipartition Theorem, every quadratic degree of freedom in the Hamiltonian contributes an average energy of $\frac{1}{2}k_B T$. Therefore, the total average energy per mode is the sum of these two contributions:

$$\langle E \rangle = \left(\frac{1}{2}k_B T\right) + \left(\frac{1}{2}k_B T\right) = k_B T \tag{3.22}$$

Substituting this result into the total energy integral:

$$u_{classical} = \int_0^\infty g(\omega)(k_B T)\, d\omega \propto \int_0^\infty \omega^2 d\omega \to \infty \tag{3.23}$$

This implies infinite energy density at high frequencies, a failure known as the **Ultraviolet Catastrophe**. This paradox is resolved by the quantum Planck distribution derived above.

3.4 Summary of Blackbody Radiation

Symbol	**Formula**
Hamiltonian H_{EM}	$H_{EM} = \sum_{\mathbf{k}} \left(\frac{\epsilon_0}{2} \dot{q}_{\mathbf{k}}^2 + \frac{\epsilon_0}{2} \omega_{\mathbf{k}}^2 q_{\mathbf{k}}^2 \right)$
Partition Func. Z_ω	$Z_\omega = \frac{1}{1 - e^{-\beta\hbar\omega}}$ (Geometric Series)
Photon DOS $g(\omega)$	$g(\omega) = \frac{V\omega^2}{\pi^2 c^3}$ $(\propto \omega^2)$
Spectral Density $u(\omega)$	$u(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1}$
Total Energy $u(T)$	$u(T) = \frac{\pi^2 k_B^4}{15\hbar^3 c^3} T^4 \propto T^4$
Stefan-Boltzmann	$J = \sigma T^4$
Classical Limit	$\langle E \rangle = k_B T$ (Rayleigh-Jeans)
UV Catastrophe	$\int_0^\infty \omega^2 (k_B T) d\omega \to \infty$

Chapter 4

Lattice Vibrations: Heat Capacity of Solids

In this chapter, we derive the heat capacity of solids. The fundamental approach is to model the solid as a collection of oscillating atoms. By finding the **normal modes** of the system, we can treat the complex coupled vibrations as a set of independent harmonic oscillators.

4.1 The Einstein Model

4.1.1 Basic Assumptions

Einstein proposed a simplified model with two key assumptions:

1. The atoms vibrate independently of each other.
2. All $3N$ vibrational modes oscillate with the **same frequency**, ω_E (the Einstein frequency).

4.1.2 Partition Function

Since the oscillators are independent, the total partition function Z for the crystal is the product of the partition functions of the individual modes (Z_k).

$$Z = \prod_{k=1}^{3N} Z_k \implies \ln Z = \sum_{k=1}^{3N} \ln Z_k = 3N \ln Z_{\text{1-mode}} \tag{4.1}$$

For a single quantum harmonic oscillator, the partition function is:

$$Z_k = \sum_{n=0}^{\infty} e^{-\beta E_n} = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega_E(n+1/2)} = \frac{e^{-\beta\hbar\omega_E/2}}{1 - e^{-\beta\hbar\omega_E}} \tag{4.2}$$

4.1.3 Internal Energy (U)

Using the thermodynamic relation $U = -\frac{\partial \ln Z}{\partial \beta}$, we calculate the total energy. If we consider the excitations as **phonons** (quasiparticles) and ignore the zero-point energy constant, the energy becomes:

$$U = \frac{3N\hbar\omega_E}{e^{\beta\hbar\omega_E} - 1} \tag{4.3}$$

This represents $3N$ modes, each occupied by a number of phonons given by the **Bose-Einstein distribution**.

4.1.4 Success and Failure

The heat capacity is $C_V = (\frac{\partial U}{\partial T})_V$.

- **High Temperature Limit ($T \to \infty$):** Expanding the exponential, $C_V \to 3Nk_B$, which recovers the classical **Dulong-Petit Law**.
- **Low Temperature Limit ($T \to 0$):** The model predicts $C_V \propto e^{-\hbar\omega_E/k_BT}$.

Critique: Experimental data shows that as $T \to 0$, $C_V \propto T^3$. The Einstein model fails at low temperatures because it ignores low-frequency sound waves (acoustic modes).

4.2 The Debye Model

To correct the Einstein model, Debye treated the solid as a continuous elastic medium, allowing for a spectrum of frequencies rather than a single ω_E.

4.2.1 Dispersion Relation

Analogous to photons, quantized sound waves are called **phonons**. In the long-wavelength limit, the dispersion relation is linear:

$$E = \hbar\omega = \hbar k c_s \tag{4.4}$$

where c_s is the speed of sound.

4.2.2 Density of States Calculation

We need to count the number of allowed modes in the frequency range ω to $\omega + d\omega$. In 3 dimensions:

$$g(\omega)d\omega = \frac{V}{2\pi^2}k^2 dk \times 3 \quad \text{(for 3 polarizations)} \tag{4.5}$$

Substituting $k = \omega/c_s$:

$$g(\omega) = \frac{3V\omega^2}{2\pi^2 c_s^3} \tag{4.6}$$

4.2.3 Constraint: The Debye Cutoff Frequency

Problem: Why must the integral equal 3N?

Question: We define the cutoff frequency ω_D using the condition:

$$\int_0^{\omega_D} g(\omega)d\omega = 3N$$

Is this an arbitrary assumption?
Answer: No, it is a necessary physical constraint imposed by the atomic nature of the solid.

- **Discrete Degrees of Freedom:** A real solid consists of N discrete atoms. Each atom has 3 degrees of freedom (x, y, z). Therefore, the system has exactly $3N$ **independent normal modes**.
- **Continuum Correction:** The density of states $g(\omega) \propto \omega^2$ assumes a continuous medium, which would imply infinite modes (up to infinite frequency).
- **The Fix:** Since the crystal is not actually continuous, we must cut off the spectrum at ω_D so that the model counts exactly $3N$ modes, matching the real degrees of freedom.

Calculation of ω_D:

$$\int_0^{\omega_D} \frac{3V\omega^2}{2\pi^2 c_s^3} d\omega = \frac{V\omega_D^3}{2\pi^2 c_s^3} = 3N \tag{4.7}$$

Solving for ω_D:

$$\omega_D = c_s \left(\frac{6\pi^2 N}{V}\right)^{1/3} \tag{4.8}$$

4.3 Thermodynamics of the Debye Solid

4.3.1 Total Energy Integral

The total energy is found by integrating the energy of each mode over the entire spectrum.

$$U = \int_0^{\omega_D} (\text{Energy of 1 phonon}) \times (\text{Occupancy}) \times (\text{Density of States}) \tag{4.9}$$

$$U = \int_0^{\omega_D} (\hbar\omega) \frac{1}{e^{\beta\hbar\omega} - 1} g(\omega) d\omega \tag{4.10}$$

Substituting $g(\omega)$:

$$U = \frac{3V\hbar}{2\pi^2 c_s^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\beta\hbar\omega} - 1} d\omega \tag{4.11}$$

4.3.2 Dimensionless Form

We introduce the dimensionless variable x:

$$x = \beta\hbar\omega = \frac{\hbar\omega}{k_B T} \quad \Longrightarrow \quad x_D = \frac{\hbar\omega_D}{k_B T} = \frac{\Theta_D}{T} \tag{4.12}$$

where Θ_D is the **Debye Temperature**. The energy becomes:

$$U = 9Nk_BT\left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^3}{e^x - 1}dx \tag{4.13}$$

4.3.3 Heat Capacity and Limits

Differentiating U with respect to T gives C_V. We focus on the two limiting cases:

1. High Temperature Limit ($T \gg \Theta_D$): The upper limit of integration is small (x is small). Expanding $e^x - 1 \approx x$:

$$\int_0^{x_D} \frac{x^3}{x}dx \approx \frac{x_D^3}{3} \tag{4.14}$$

This leads to $C_V \approx 3Nk_B$, recovering the Dulong-Petit law.

2. Low Temperature Limit ($T \ll \Theta_D$): The upper limit $\Theta_D/T \to \infty$. The integral converges to a constant value ($\frac{\pi^4}{15}$).

$$C_V \approx \frac{12\pi^4}{5}Nk_B\left(\frac{T}{\Theta_D}\right)^3 \tag{4.15}$$

Conclusion: The Debye model correctly predicts the T^3 dependence of heat capacity at low temperatures, solving the problem faced by the Einstein model.

4.4 Summary of Heat Capacity Models

Symbol	**Formula**
Einstein $\ln Z$	$\ln Z = 3N \ln\left(\frac{e^{-\beta\hbar\omega_E/2}}{1-e^{-\beta\hbar\omega_E}}\right)$
Einstein Energy U_E	$U = \frac{3N\hbar\omega_E}{e^{\beta\hbar\omega_E}-1}$
Einstein C_V (Low T)	$C_V \propto e^{-\hbar\omega_E/k_BT}$ (Exponential decay)
Phonon Dispersion	$E = \hbar\omega = \hbar k c_s$
Debye DOS $g(\omega)$	$g(\omega) = \frac{3V\omega^2}{2\pi^2 c_s^3}$ $(\propto \omega^2)$
Cutoff Freq. ω_D	$\omega_D = c_s\left(\frac{6\pi^2 N}{V}\right)^{1/3}$
Debye Energy U_D	$U = 9Nk_BT\left(\frac{T}{\Theta_D}\right)^3 \int_0^{\Theta_D/T} \frac{x^3}{e^x-1}dx$
Debye C_V (Low T)	$C_V \approx \frac{12\pi^4}{5}Nk_B\left(\frac{T}{\Theta_D}\right)^3$ $(\propto T^3)$

Chapter 5

Ultrarelativistic Ideal Gases

5.1 Introduction: The Ultrarelativistic Limit

We consider a gas of particles with non-zero mass in the ultrarelativistic limit. The general relativistic dispersion relation connects energy E and momentum p:

$$E^2 = p^2c^2 + m^2c^4 \tag{5.1}$$

There are two limiting cases:

1. **Non-relativistic ($p \ll mc$):** The energy approximates to $E \approx mc^2 + \frac{p^2}{2m}$.
2. **Ultrarelativistic ($p \gg mc$):** The mass term becomes negligible, and the relation simplifies to:
$$E = pc \tag{5.2}$$

We will focus on the second case (Ultrarelativistic) to derive the thermodynamic properties.

5.2 The Single-Particle Partition Function (Z_1)

For a continuous spectrum, the single-particle partition function is an integral over the density of states $g(k)$:

$$Z_1 = \int_0^\infty e^{-\beta E} g(k)\, dk = \int_0^\infty e^{-\beta\hbar kc} \frac{Vk^2}{2\pi^2}\, dk \tag{5.3}$$

where we used $E = \hbar kc$ and $g(k) = \frac{Vk^2}{2\pi^2}$ (accounting for spin/degeneracy if applicable, here implied in the constant).

5.2.1 Solving the Integral

We perform a substitution to solve the integral. Let $x = \beta\hbar ck$. Then $k = \frac{x}{\beta\hbar c}$ and $dk = \frac{dx}{\beta\hbar c}$. Substituting these into the integral:

$$Z_1 = \frac{V}{2\pi^2}\left(\frac{1}{\beta\hbar c}\right)^3 \int_0^\infty e^{-x} x^2\, dx \tag{5.4}$$

Using the standard integral $\int_0^\infty e^{-x}x^2dx = 2! = 2$:

$$Z_1 = \frac{V}{\pi^2}\left(\frac{k_BT}{\hbar c}\right)^3 \tag{5.5}$$

We define the thermal wavelength analog Λ for the ultrarelativistic case such that $Z_1 = \frac{V}{\Lambda^3}$. This yields:

$$\Lambda = \frac{\hbar c\pi^{2/3}}{k_BT} \tag{5.6}$$

5.3 Thermodynamics of N Indistinguishable Particles

For an ideal gas of N indistinguishable particles, the total partition function is:

$$Z_N = \frac{Z_1^N}{N!} \tag{5.7}$$

Taking the natural logarithm and using Stirling's approximation ($\ln N! \approx N\ln N - N$):

$$\ln Z_N = N\ln Z_1 - \ln N! \approx N\ln\left(\frac{V}{\Lambda^3}\right) - (N\ln N - N) \tag{5.8}$$

Note that since $\Lambda \propto T^{-1}$ (unlike $T^{-1/2}$ in the non-relativistic case), $\ln Z_N$ contains a term $3N\ln T$.

5.3.1 Derivation of Thermodynamic Potentials

1. Internal Energy (U)

$$U = -\frac{\partial \ln Z_N}{\partial\beta} = 3Nk_BT \tag{5.9}$$

(Compared to $\frac{3}{2}Nk_BT$ for non-relativistic gas).

2. Heat Capacity (C_V)

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = 3Nk_B \tag{5.10}$$

3. Helmholtz Free Energy (F)

$$F = -k_BT\ln Z_N = Nk_BT\left[\ln(n\Lambda^3) - 1\right] \tag{5.11}$$

4. Pressure (P)

$$P = -\left(\frac{\partial F}{\partial V}\right)_T = \frac{Nk_BT}{V} = nk_BT \tag{5.12}$$

Interestingly, the relationship between pressure and energy density $u = U/V$ is:

$$P = \frac{3Nk_BT}{3V} = \frac{U}{3V} = \frac{u}{3} \tag{5.13}$$

This differs from the non-relativistic case where $P = \frac{2}{3}u$.

5. Enthalpy (H)

$$H = U + PV = 3Nk_BT + Nk_BT = 4Nk_BT \tag{5.14}$$

6. Entropy (S) and Gibbs Free Energy (G) Using $G = H - TS$ and the previous results:

$$S = Nk_B[4 - \ln(n\Lambda^3)] \tag{5.15}$$
$$G = Nk_BT\ln(n\Lambda^3) \tag{5.16}$$

5.4 Conceptual Problem: Adiabatic Expansion

Problem: In the adiabatic expansion of an ultrarelativistic gas, we stated that $dS = 0$. We know from the Second Law of Thermodynamics that for a thermally isolated system, $dS \geq 0$. Why do we assume $dS = 0$ here?

Solution: The condition $dS = 0$ implies that the expansion is not just adiabatic (isolated), but also **Reversible** (Quasi-static).

- **General Case:** For any adiabatic process, $dQ = 0$. The Second Law states $dS \geq 0$. If the expansion is fast or uncontrolled (like a free expansion), entropy strictly increases ($dS > 0$).
- **Our Case:** To derive specific state equations like $PV^\gamma = \text{const}$, we must assume the system stays in equilibrium throughout the process. This requires the process to be reversible.
- **Conclusion:** Using the definition $dS = dQ_{\text{rev}}/T$, and since $dQ_{\text{rev}} = 0$, we get $dS = 0$. This is often called an *Isentropic Expansion.*

5.4.1 Adiabatic Relations for Ultrarelativistic Gas

Since S is constant, the term inside the log of the entropy equation must be constant:

$$n\Lambda^3 = \text{constant} \implies \frac{N}{V}\left(\frac{1}{T}\right)^3 = \text{constant} \tag{5.17}$$

Rearranging this yields the adiabatic relation for volume and temperature:

$$VT^3 = \text{constant} \tag{5.18}$$

Using the ideal gas law $P \propto T/V$, we can substitute $T \propto PV$ to find the pressure-volume relation:

$$PV^{4/3} = \text{constant} \tag{5.19}$$

This confirms that the adiabatic index $\gamma = 4/3$ for an ultrarelativistic gas (compared to $\gamma = 5/3$ for the non-relativistic case).

5.5 Summary: Non-Relativistic vs. Ultrarelativistic

Property	**Non-Relativistic**	**Ultrarelativistic**
Dispersion	$E \approx p^2/2m$	$E = pc$
Partition Function (Z_1)	$\frac{V}{\lambda_{th}^3}$	$\frac{V}{\Lambda^3}$
Wavelength Definition	$\lambda_{th} = \frac{h}{\sqrt{2\pi m k_B T}}$	$\Lambda = \frac{\hbar c \pi^{2/3}}{k_B T}$
Internal Energy (U)	$\frac{3}{2} N k_B T$	$3 N k_B T$
Enthalpy (H)	$\frac{5}{2} N k_B T$	$4 N k_B T$
Pressure Relation	$P = \frac{2}{3} u$	$P = \frac{1}{3} u$
Entropy (S)	$N k_B [\frac{5}{2} - \ln(n \lambda_{th}^3)]$	$N k_B [4 - \ln(n \Lambda^3)]$
Adiabatic Law	$PV^{5/3} = \text{const}$	$PV^{4/3} = \text{const}$

Chapter 6

Interacting Gases: Mayer Cluster Expansion

The goal of this lecture note is to derive the Van der Waals equation of state from the canonical partition function using the Mayer Cluster Expansion method. We begin with the fundamental statistical mechanics definitions and apply approximations suitable for a dilute gas.

6.1 The Canonical Partition Function

For a system of N indistinguishable particles in a volume V at temperature T, the classical partition function $Z(N, V, T)$ is given by the integral over phase space:

$$Z(N, V, T) = \frac{1}{N! h^{3N}} \int \prod_{i=1}^{N} d^3 p_i d^3 r_i \, e^{-\beta H} \tag{6.1}$$

The Hamiltonian H is the sum of the kinetic energy and the pairwise potential energy $U(r_{jk})$:

$$H = \sum_{j=1}^{N} \frac{p_j^2}{2m} + \sum_{j<k} U(r_{jk}) \tag{6.2}$$

6.1.1 Separation of Kinetic and Potential Terms

The kinetic energy term depends only on momentum (p), while the potential energy depends only on position (r). Therefore, we can separate the integrals. The momentum integral yields the thermal de Broglie wavelength, $\lambda = \sqrt{\frac{h^2}{2\pi m k_B T}}$.

Thus, the partition function simplifies to a configuration integral over positions:

$$Z = \frac{1}{N! \lambda^{3N}} \int \prod_{i=1}^{N} d^3 r_i \, e^{-\beta \sum_{j<k} U(r_{jk})} \tag{6.3}$$

6.2 The Mayer Cluster Expansion

To handle the interacting potential term, we introduce the **Mayer f-function**:

$$f_{ij} = e^{-\beta U(r_{ij})} - 1 \tag{6.4}$$

This function represents the deviation from ideal gas behavior. It is zero when particles are far apart ($U \approx 0$) and non-zero when they interact. Using this, the Boltzmann factor becomes a product:

$$e^{-\beta \sum_{j<k} U(r_{jk})} = \prod_{j<k} (1 + f_{jk}) \tag{6.5}$$

6.2.1 Low Density Approximation

For a dilute gas, simultaneous collisions of three or more particles are negligible. We approximate the expansion by keeping only the terms involving single pairs (linear in f_{jk}):

$$\prod_{j<k} (1 + f_{jk}) \approx 1 + \sum_{j<k} f_{jk} \tag{6.6}$$

Substituting this back into the partition function integral:

$$Z \approx \frac{1}{N! \lambda^{3N}} \int \prod_{i=1}^{N} d^3 r_i \left(1 + \sum_{j<k} f_{jk} \right) \tag{6.7}$$

6.3 The Pair Counting Factor

Question: Why does the sum $\sum_{j<k} f_{jk}$ turn into a single term multiplied by $\frac{N^2}{2}$?

Solution: The integral involves a sum over all possible pairs of particles. However, since all particles are identical, every pair (j, k) contributes the exact same value to the integral.

1. We replace the sum $\sum_{j<k}$ with a single representative term (e.g., pair 1-2) multiplied by the total number of distinct pairs.

2. The number of ways to choose 2 particles from N is:

$$\binom{N}{2} = \frac{N(N-1)}{2} \approx \frac{N^2}{2} \quad \text{(for large } N\text{)} \tag{6.8}$$

3. Therefore, we simplify the integration:

$$\int \left(\sum_{j<k} f_{jk} \right) \cdots \approx \frac{N^2}{2} \int f_{12} \cdots \tag{6.9}$$

6.4 Detailed Derivation: Coordinate Transformation

We need to evaluate the integral involving the interacting pair (Particle 1 and Particle 2). The term of interest is:

$$I_{12} = \int \int d^3 r_1 \, d^3 r_2 \, f(|\vec{r}_1 - \vec{r}_2|) \tag{6.10}$$

To solve this, we decouple the relative motion from the center of mass motion using a change of variables.

6.4.1 Definition of New Coordinates

We define the Center of Mass vector ($\vec{R}$) and the Relative Separation vector ($\vec{r}$):

$$\vec{R} = \frac{\vec{r}_1 + \vec{r}_2}{2} \tag{6.11}$$

$$\vec{r} = \vec{r}_1 - \vec{r}_2 \tag{6.12}$$

Inverting these relations allows us to express the original coordinates in terms of the new ones:

$$\vec{r}_1 = \vec{R} + \frac{\vec{r}}{2} \tag{6.13}$$

$$\vec{r}_2 = \vec{R} - \frac{\vec{r}}{2} \tag{6.14}$$

6.4.2 The Jacobian Determinant

When changing integration variables, we must include the Jacobian determinant (J) to preserve the volume element:

$$d^3r_1 d^3r_2 = |J|\, d^3R\, d^3r \tag{6.15}$$

The Jacobian matrix for the x-components is:

$$J_x = \det \begin{pmatrix} \frac{\partial x_1}{\partial X} & \frac{\partial x_1}{\partial x} \\ \\ \frac{\partial x_2}{\partial X} & \frac{\partial x_2}{\partial x} \end{pmatrix} = \det \begin{pmatrix} 1 & \frac{1}{2} \\ \\ 1 & -\frac{1}{2} \end{pmatrix} \tag{6.16}$$

Calculating the determinant:

$$J_x = (1)(-\frac{1}{2}) - (1)(\frac{1}{2}) = -1 \tag{6.17}$$

Since this transformation applies identically to the y and z components, the total Jacobian is the product of the three dimensions:

$$J_{\text{total}} = J_x J_y J_z = (-1)(-1)(-1) = -1 \tag{6.18}$$

We take the absolute value for the volume element, so $|J| = 1$. Thus, the integration measure transforms simply as:

$$d^3r_1 d^3r_2 = d^3R\, d^3r \tag{6.19}$$

6.4.3 Separating the Integrals

Substituting the variables into the integral I_{12}:

$$I_{12} = \int_V d^3R \int d^3r\, f(r) \tag{6.20}$$

Because the interaction $f(r)$ depends **only** on the relative distance $|\vec{r}|$ and not on the absolute position $\vec{R}$, the integrals separate completely:

1. **Center of Mass Integral:** The center of mass $\vec{R}$ can be anywhere inside the volume V:

$$\int_V d^3R = V \tag{6.21}$$

2. **Relative Integral:** This remains as the interaction term:

$$\int d^3r\, f(r) \tag{6.22}$$

6.4.4 Including Spectator Particles

The original partition function integral included N particles. We have handled particles 1 and 2. The remaining $N-2$ particles (3, 4, ..., N) are non-interacting "spectators" in this term. Their integrals simply yield the volume:

$$\prod_{i=3}^{N} \int d^3r_i = V^{N-2} \tag{6.23}$$

6.4.5 Final Result for the Term

Multiplying the pieces together (Spectators $\times$ Center of Mass $\times$ Relative Interaction):

$$\text{Total Term} = (V^{N-2}) \times (V) \times \left(\int d^3r f(r) \right) = V^{N-1} \int d^3r f(r) \tag{6.24}$$

6.5 Thermodynamics and Equation of State

The Helmholtz Free Energy is given by $F = -k_B T \ln Z$.

$$F = F_{\text{ideal}} - N k_B T \ln \left(1 + \frac{N}{2V} \int d^3r f(r) \right) \tag{6.25}$$

Using $\ln(1+x) \approx x$:

$$F \approx F_{\text{ideal}} - \frac{N^2 k_B T}{2V} \int d^3r f(r) \tag{6.26}$$

We compute the pressure $P = -(\partial F/\partial V)_T$:

$$P = \frac{N k_B T}{V} \left(1 - \frac{N}{2V} \int d^3r f(r) \right) \tag{6.27}$$

6.6 Real Potentials and Van der Waals Constants

We model the potential $U(r)$ with a hard-sphere repulsion and weak attraction:

$$U(r) = \begin{cases} \infty & r < r_0 \quad \text{(Hard Core)} \\ -U_0 (r_0/r)^6 & r \geq r_0 \quad \text{(Attraction)} \end{cases} \tag{6.28}$$

Splitting the integral $\int f(r) d^3r$ into two parts:

1. **Repulsive Part (0 to r_0):** Here $f(r) = -1$.
2. **Attractive Part (r_0 to ∞):** Here $f(r) \approx -\beta U(r)$.

This leads to the Van der Waals constants:

- $b = \frac{2}{3}\pi r_0^3$ (related to excluded volume)
- $a = \frac{2}{3}\pi r_0^3 U_0$ (related to attractive strength)

Finally, substituting these back gives the **Van der Waals Equation**:

$$\left(P + \frac{N^2}{V^2}a\right)(V - Nb) = Nk_BT \tag{6.29}$$

Chapter 7

Real Gas Thermodynamics: Expansion & Cooling

7.1 Joule Expansion (Free Expansion)

The Joule expansion describes a gas expanding adiabatically into a vacuum. Since no work is done ($W = 0$) and no heat is exchanged ($Q = 0$), the **internal energy remains constant** ($U = \text{constant}$).

7.1.1 The Joule Coefficient

We quantify the temperature change during this expansion using the **Joule coefficient** μ_J:

$$\mu_J = \left(\frac{\partial T}{\partial V}\right)_U \tag{7.1}$$

To calculate this, we use the cyclic chain rule (triple product rule) involving $T, V,$ and U:

$$\left(\frac{\partial T}{\partial V}\right)_U \left(\frac{\partial V}{\partial U}\right)_T \left(\frac{\partial U}{\partial T}\right)_V = -1 \tag{7.2}$$

Rearranging for μ_J:

$$\mu_J = -\frac{(\frac{\partial U}{\partial V})_T}{(\frac{\partial U}{\partial T})_V} = -\frac{1}{C_V}\left(\frac{\partial U}{\partial V}\right)_T \tag{7.3}$$

Using the thermodynamic identity derived from the fundamental relation $dU = TdS - PdV$ and Maxwell relations:

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P \tag{7.4}$$

Substituting this back gives the general formula:

$$\mu_J = -\frac{1}{C_V}\left[T\left(\frac{\partial P}{\partial T}\right)_V - P\right] \tag{7.5}$$

7.1.2 Case 1: Ideal Gas

For an ideal gas, the equation of state is $P = \frac{RT}{V}$. The derivative is:

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V} \tag{7.6}$$

Substituting into the bracket term:

$$T\left(\frac{R}{V}\right) - P = P - P = 0 \tag{7.7}$$

Conclusion: $\mu_J = 0$. An ideal gas does not change temperature during a free expansion ($dT = 0$).

7.1.3 Case 2: Van der Waals Gas

For a real gas obeying the Van der Waals equation:

$$P = \frac{RT}{V-b} - \frac{a}{V^2} \tag{7.8}$$

The derivative is:

$$\left(\frac{\partial P}{\partial T}\right)_V = \frac{R}{V-b} \tag{7.9}$$

Substituting this into the general formula:

$$T\left(\frac{\partial P}{\partial T}\right)_V - P = \frac{RT}{V-b} - \left(\frac{RT}{V-b} - \frac{a}{V^2}\right) = \frac{a}{V^2} \tag{7.10}$$

Thus, the Joule coefficient is non-zero:

$$\mu_J = -\frac{1}{C_V}\frac{a}{V^2} \tag{7.11}$$

Temperature Change calculation

To find the total temperature change ΔT when expanding from V_1 to V_2:

$$\Delta T = \int_{V_1}^{V_2} \mu_J \, dV = -\frac{a}{C_V}\int_{V_1}^{V_2} \frac{dV}{V^2} \tag{7.12}$$

$$\Delta T = -\frac{a}{C_V}\left[-\frac{1}{V}\right]_{V_1}^{V_2} = -\frac{a}{C_V}\left(\frac{1}{V_1} - \frac{1}{V_2}\right) \tag{7.13}$$

Since the gas expands ($V_2 > V_1$), the term in parentheses is positive. Therefore, $\Delta T < 0$.
Conclusion: A real gas **cools down** during free expansion.

Physical Intuition: Why does it cool?

The Physics of Cooling

Why does $\Delta T < 0$?

- **Potential Energy:** In a real gas, particles attract each other (parameter a). The potential energy $U(r)$ is negative and increases (approaches zero) as the distance $\langle d \rangle$ between particles increases.
- **Expansion:** As volume increases ($V \uparrow$), the average distance between particles increases ($\langle d \rangle \uparrow$). This requires work against the attractive forces, so the **Potential Energy increases**.
- **Energy Conservation:** Since total internal energy $U = K.E. + P.E.$ is constant in a Joule expansion, if P.E. goes up, Kinetic Energy (K.E.) must go down.
- **Temperature:** Since $T \propto$ Average K.E., the temperature drops.

7.2 Isothermal Expansion

We now consider an expansion where temperature is held constant ($dT = 0$).

7.2.1 Change in Internal Energy (ΔU)

Using the thermodynamic relation derived earlier:

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial P}{\partial T}\right)_V - P \tag{7.14}$$

For a Van der Waals gas, we found the right-hand side equals $\frac{a}{V^2}$. Thus:

$$\Delta U = \int_{V_1}^{V_2} \left(\frac{\partial U}{\partial V}\right)_T dV = \int_{V_1}^{V_2} \frac{a}{V^2} dV = a\left(\frac{1}{V_1} - \frac{1}{V_2}\right) \tag{7.15}$$

This confirms that internal energy depends on volume for a real gas (unlike an ideal gas).

7.2.2 Change in Entropy (ΔS)

We start with S as a function of T and V:

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV \tag{7.16}$$

Using the definition of heat capacity $C_V = T(\partial S/\partial T)_V$ and the Maxwell relation $(\partial S/\partial V)_T = (\partial P/\partial T)_V$:

$$dS = \frac{C_V}{T} dT + \left(\frac{\partial P}{\partial T}\right)_V dV \tag{7.17}$$

Integrating for a Van der Waals gas where $(\frac{\partial P}{\partial T})_V = \frac{R}{V-b}$:

$$S = \int \frac{C_V}{T} dT + \int \frac{R}{V-b} dV \tag{7.18}$$

$$S = C_V \ln T + R \ln(V-b) + \text{constant} \tag{7.19}$$

7.3 Joule-Kelvin Expansion (Throttling Process)

This process involves forcing a gas through a porous plug from a high pressure region to a low pressure region.

7.3.1 Conservation of Enthalpy

Unlike Joule expansion (constant U), the Joule-Kelvin expansion is an **isenthalpic process**:

$$H_1 = H_2 \implies dH = 0 \tag{7.20}$$

7.3.2 The Joule-Kelvin Coefficient

We are interested in how temperature changes with pressure:

$$\mu_{JK} = \left(\frac{\partial T}{\partial P}\right)_H \tag{7.21}$$

Using the cyclic chain rule for T, P, H:

$$\mu_{JK} = -\frac{(\frac{\partial H}{\partial P})_T}{(\frac{\partial H}{\partial T})_P} = -\frac{1}{C_P}\left(\frac{\partial H}{\partial P}\right)_T \tag{7.22}$$

Using the fundamental relation $dH = TdS + VdP$:

$$\left(\frac{\partial H}{\partial P}\right)_T = T\left(\frac{\partial S}{\partial P}\right)_T + V \tag{7.23}$$

Applying the Maxwell relation $\left(\frac{\partial S}{\partial P}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_P$:

$$\left(\frac{\partial H}{\partial P}\right)_T = -T\left(\frac{\partial V}{\partial T}\right)_P + V \tag{7.24}$$

Substituting this back gives the general formula:

$$\mu_{JK} = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right] \tag{7.25}$$

7.3.3 Case 1: Ideal Gas

For an ideal gas, $V = \frac{RT}{P}$. The derivative is $(\frac{\partial V}{\partial T})_P = \frac{R}{P}$.

$$T\left(\frac{R}{P}\right) - V = V - V = 0 \tag{7.26}$$

Result: $\mu_{JK} = 0$. An ideal gas does not cool upon throttling.

7.3.4 Entropy and Irreversibility

Even though $dH = 0$ and for an ideal gas $dT = 0$, the process is **irreversible**. Since $dH = TdS + VdP = 0$, we have:

$$dS = -\frac{V}{T}dP \tag{7.27}$$

For an ideal gas expansion ($P_1 > P_2$), ΔS:

$$\Delta S = -\int_{P_1}^{P_2} \frac{nR}{P}dP = nR\ln\left(\frac{P_1}{P_2}\right) \tag{7.28}$$

Since $P_1 > P_2$, the logarithm is positive, so $\Delta S > 0$. The entropy increases, confirming the process is irreversible.

7.3.5 Inversion Condition

For real gases, cooling ($\mu_{JK} > 0$) or heating ($\mu_{JK} < 0$) depends on the temperature. The transition point is called the **Inversion Temperature**, defined by $\mu_{JK} = 0$:

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{V}{T} \tag{7.29}$$

Below this temperature, the gas cools upon expansion; above it, the gas heats up.

7.4 Summary of Real Gas Expansions

Symbol	**Formula**
Joule Expansion	$U = \text{constant} \quad (dU = 0)$
Joule Coeff. μ_J	$\mu_J = \left(\frac{\partial T}{\partial V}\right)_U = -\frac{1}{C_V}\left[T\left(\frac{\partial P}{\partial T}\right)_V - P\right]$
μ_J (VdW Gas)	$\mu_J = -\frac{a}{C_V V^2}$
ΔT (VdW Joule)	$\Delta T = -\frac{a}{C_V}\left(\frac{1}{V_1} - \frac{1}{V_2}\right)$
ΔU_T (Isothermal)	$\Delta U_T = a\left(\frac{1}{V_1} - \frac{1}{V_2}\right)$
Entropy S_{VdW}	$S = C_V \ln T + R\ln(V - b) + \text{const}$
Joule-Kelvin	$H = \text{constant} \quad (dH = 0)$
J-K Coeff. μ_{JK}	$\mu_{JK} = \left(\frac{\partial T}{\partial P}\right)_H = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right]$
Inversion Temp.	$\mu_{JK} = 0 \implies \left(\frac{\partial V}{\partial T}\right)_P = \frac{V}{T}$

Chapter 8

Phase Transitions and The Ising Model

A phase transition is the transformation of a thermodynamic system from one phase or state of matter to another.

- **Latent Heat (L):** The heat absorbed or released during a phase change at constant temperature.
$$L = \Delta Q_{\text{rev}} = T_c(S_2 - S_1) \tag{8.1}$$
This implies a discontinuity in entropy at the transition point.

- **Equilibrium Condition:** For two phases to be in equilibrium, their chemical potentials must be equal.
$$dG_{\text{tot}} = \mu_1 dN_1 + \mu_2 dN_2 = 0 \implies \mu_1 = \mu_2 \tag{8.2}$$

8.1 The Clausius-Clapeyron Equation

We derive the equation describing the slope of the phase boundary in the $P - T$ plane. Consider two phases (1 and 2) in equilibrium along a boundary curve.

$$\mu_1(p, T) = \mu_2(p, T) \tag{8.3}$$

If we move slightly along the boundary to $(p + dp, T + dT)$, equality must hold:

$$d\mu_1 = d\mu_2 \tag{8.4}$$

Using the fundamental relation $d\mu = -sdT + vdp$ (where s and v are per-particle entropy and volume):
$$-s_1 dT + v_1 dp = -s_2 dT + v_2 dp \tag{8.5}$$

Rearranging terms:
$$(v_2 - v_1)dp = (s_2 - s_1)dT \tag{8.6}$$

Since the latent heat per particle is $l = T(s_2 - s_1)$, we arrive at the **Clausius-Clapeyron Equation**:
$$\frac{dp}{dT} = \frac{s_2 - s_1}{v_2 - v_1} = \frac{L}{T(V_2 - V_1)} \tag{8.7}$$

8.2 Temperature Dependence of Latent Heat

Problem: Evaluate the temperature dependence of the latent heat along the phase boundary and find the phase boundary equation.

Step 1: The Total Derivative along the Boundary We are moving along the phase boundary $p(T)$. The rate of change of any quantity $X(T, p)$ along this line is:

$$\frac{d}{dT} = \left(\frac{\partial}{\partial T}\right)_p + \frac{dp}{dT}\left(\frac{\partial}{\partial p}\right)_T \tag{8.8}$$

Step 2: Differentiating Entropy Difference We know $\Delta S = \frac{L}{T}$. Let's differentiate both sides with respect to T along the boundary:

$$\frac{d}{dT}\left(\frac{L}{T}\right) = \frac{d}{dT}(S_{\text{vap}} - S_{\text{liq}}) \tag{8.9}$$

Using the operator definition from Step 1 on the RHS:

$$\frac{1}{T}\frac{dL}{dT} - \frac{L}{T^2} = \left(\frac{\partial(S_v - S_l)}{\partial T}\right)_p + \frac{dp}{dT}\left(\frac{\partial(S_v - S_l)}{\partial p}\right)_T \tag{8.10}$$

Using Kirchhoff's relation logic and neglecting liquid volume, we find:

$$\frac{dL}{dT} = C_{p,v} - C_{p,l} \tag{8.11}$$

8.3 Stability and Metastability

Understanding Phase Stability

Question: What is the physical meaning of the derivatives of the chemical potential?

$$\left(\frac{\partial \mu}{\partial p}\right)_T = v \quad \text{and} \quad \left(\frac{\partial \mu}{\partial T}\right)_p = -s$$

Explanation: The chemical potential μ is the Gibbs free energy per particle. Systems evolve to **minimize** μ. These derivatives tell us how stability changes with conditions:

1. **Pressure Dependence ($\partial\mu/\partial p = v$):**
   - The slope of μ vs p is the molecular volume v.
   - Since $v_{\text{gas}} \gg v_{\text{liquid}}$, the gas chemical potential rises much faster with pressure than the liquid.
   - **Result:** At high pressures, the phase with the **smaller volume** (liquid/solid) becomes more stable.
2. **Temperature Dependence ($\partial\mu/\partial T = -s$):**
   - The slope of μ vs T is negative entropy $-s$.
   - Since $s_{\text{gas}} \gg s_{\text{liquid}}$, the gas curve slopes downward much steeper than the liquid.
   - **Result:** At high temperatures, the phase with the **higher entropy** (gas) becomes stable.

8.4 The Ising Model

The Ising model describes magnetic phase transitions. We consider spins $s_i = \pm 1$ on a lattice.

$$H = -J \sum_{\langle ij \rangle} s_i s_j - B \sum_i s_i \tag{8.12}$$

where $\langle ij \rangle$ denotes nearest-neighbor pairs and J is the exchange energy.

8.4.1 Derivation of Mean Field Energy

This derivation relies on approximating the interaction term and summing over the lattice using the coordination number q.

Step 1: Approximate the Interaction Term

We expand the product of two spins $s_i s_j$ relative to the average magnetization m:

$$s_i s_j = (s_i - m)(s_j - m) + m(s_j - m) + m(s_i - m) + m^2 \tag{8.13}$$

We assume fluctuations $(s_i - m)(s_j - m)$ are negligible. Removing this term simplifies the expression:

$$s_i s_j \approx m(s_j - m) + m(s_i - m) + m^2 = m(s_i + s_j) - m^2 \tag{8.14}$$

Step 2: Substitute into the Hamiltonian
Substituting this approximation into the energy sum:

$$E_{\text{mf}} = -J \sum_{\langle ij \rangle} [m(s_i + s_j) - m^2] - B \sum_i s_i \tag{8.15}$$

Step 3: Evaluate the Sums
To simplify, we convert the sum over pairs into a sum over sites using the coordination number q .

- **Constant term:** The term $\sum_{\langle ij \rangle}(-m^2)$ sums a constant over all $\frac{1}{2}Nq$ pairs:

$$-J \sum_{\langle ij \rangle}(-m^2) = \frac{1}{2} JNqm^2 \tag{8.16}$$

- **Spin term:** In the sum over pairs, every spin s_i appears q times:

$$-Jm \sum_{\langle ij \rangle}(s_i + s_j) = -Jqm \sum_i s_i \tag{8.17}$$

Step 4: Combine and Factor
Combining these results yields the Mean Field Energy Equation (5.16):

$$E_{\text{mf}} = \frac{1}{2} JNqm^2 - (Jqm + B) \sum_i s_i \tag{8.18}$$

8.4.2 Derivation of the Partition Function (Z)

Using the mean field energy derived above, we calculate Z.
Step 1: Definition

$$Z = \sum_{\{s_i\}} e^{-\beta E_{\text{mf}}} \tag{8.19}$$

Step 2: Effective Field Substitution
Define the effective field $B_{\text{eff}} = Jqm + B$. Substituting E_{mf}:

$$Z = \sum_{\{s_i\}} \exp\left[-\beta \left(\frac{1}{2} JNqm^2 - B_{\text{eff}} \sum_i s_i\right)\right] \tag{8.20}$$

Step 3: Factor out Constants

$$Z = \exp\left(-\frac{1}{2}\beta JNqm^2\right) \sum_{\{s_i\}} \exp\left(\beta B_{\text{eff}} \sum_i s_i\right) \tag{8.21}$$

Step 4: Factorize the Sum
Since spins are independent in the effective field, the sum over configurations becomes a product of individual sums:

$$\sum_{\{s_i\}} \prod_{i=1}^{N} e^{\beta B_{\text{eff}} s_i} = \left(\sum_{s_1=\pm 1} e^{\beta B_{\text{eff}} s_1}\right) \times \cdots \times \left(\sum_{s_N=\pm 1} e^{\beta B_{\text{eff}} s_N}\right)$$

$$\sum_{\{s_i\}} \prod_{i=1}^{N} e^{\beta B_{\text{eff}} s_i} = \left[\sum_{s=\pm 1} e^{\beta B_{\text{eff}} s} \right]^N \tag{8.22}$$

then we have:

$$\sum_{s=\pm 1} e^{\beta B_{\text{eff}} s} = e^{\beta B_{\text{eff}}} + e^{-\beta B_{\text{eff}}} = 2\cosh(\beta B_{\text{eff}}) \tag{8.23}$$

$$Z = e^{-\frac{1}{2}\beta J N q m^2} \left[2\cosh(\beta B_{\text{eff}})\right]^N \tag{8.24}$$

8.4.3 Calculation of Magnetization (m)

The magnetization is defined as the average spin per site.

Formula:

$$m = \frac{1}{N} \sum_i \langle s_i \rangle = \frac{1}{N\beta} \frac{\partial \ln Z}{\partial B} \tag{8.25}$$

Derivation: Taking the logarithm of Z:

$$\ln Z = -\frac{1}{2}\beta J N q m^2 + N \ln 2 + N \ln \cosh(\beta(B + Jqm)) \tag{8.26}$$

Differentiating with respect to B:

$$\frac{\partial \ln Z}{\partial B} = 0 + 0 + N \frac{1}{\cosh(\beta B_{\text{eff}})} \sinh(\beta B_{\text{eff}}) \cdot \beta \tag{8.27}$$

$$\frac{\partial \ln Z}{\partial B} = N\beta \tanh(\beta B_{\text{eff}}) \tag{8.28}$$

Substituting back into the formula for m:

$$m = \frac{1}{N\beta}(N\beta \tanh(\beta B_{\text{eff}})) \tag{8.29}$$

$$m = \tanh(\beta(B + Jqm)) \tag{8.30}$$

This is the self-consistency equation for the magnetization.

8.5 Summary of Phase Transitions and Ising Model

Symbol	**Formula**
Latent Heat L	$L = T(S_2 - S_1) = T\Delta S$
Clausius-Clapeyron	$\dfrac{dp}{dT} = \dfrac{L}{T(V_2 - V_1)} = \dfrac{\Delta s}{\Delta v}$
Stability Derivs.	$\left(\dfrac{\partial \mu}{\partial T}\right)_p = -s, \quad \left(\dfrac{\partial \mu}{\partial p}\right)_T = v$
Ising Hamiltonian	$H = -J \sum_{\langle ij \rangle} s_i s_j - B \sum_i s_i$
Mean Field Energy	$E_{\text{mf}} = \dfrac{1}{2} JNqm^2 - (Jqm + B) \sum_i s_i$
Partition Function Z	$Z = e^{-\frac{1}{2}\beta JNqm^2} \left[2\cosh(\beta(B + Jqm))\right]^N$
Magnetization m	$m = \tanh(\beta(B + Jqm))$

Chapter 9

Bose–Einstein and Fermi–Dirac Distributions

9.1 Identical Particles in Quantum Mechanics

In classical mechanics, particles are distinguishable. In quantum mechanics, identical particles are indistinguishable, which imposes symmetry constraints on the wavefunction.

Consider the exchange operator $\hat{P}_{12}$ which swaps particle 1 and particle 2:

$$\hat{P}_{12}\Psi(\vec{r}_1, \vec{r}_2) = \Psi(\vec{r}_2, \vec{r}_1) \tag{9.1}$$

For identical particles, the Hamiltonian $\hat{H}$ is invariant under particle exchange, meaning they commute:

$$[\hat{H}, \hat{P}_{12}] = 0 \tag{9.2}$$

This implies that the eigenstates of $\hat{H}$ are also eigenstates of $\hat{P}_{12}$. Since applying the exchange twice returns the system to the original state ($\hat{P}_{12}^2 = I$), the eigenvalues must be $\lambda^2 = 1$, so $\lambda = \pm 1$.

$$\Psi(\vec{r}_2, \vec{r}_1) = \pm\Psi(\vec{r}_1, \vec{r}_2) \tag{9.3}$$

- **Bosons (+):** Symmetric wavefunctions (integer spin).
- **Fermions (−):** Antisymmetric wavefunctions (half-integer spin).

9.2 The Grand Canonical Ensemble

To calculate the statistics of these particles, we utilize the **Grand Canonical Ensemble** (fixed μ, V, T). This avoids the mathematical difficulty of the fixed-particle constraint ($\sum n_r = N$) present in the Canonical Ensemble.

9.2.1 Partition Functions

The Grand Canonical Partition Function $\mathcal{Z}$ factorizes into a product of single-state partition functions:

$$\mathcal{Z} = \prod_r \mathcal{Z}_r \tag{9.4}$$

Depending on the particle statistics, the single-state partition function takes the form:

- **Fermi-Dirac:** $\mathcal{Z}_r = 1 + e^{-\beta(E_r-\mu)}$
- **Bose-Einstein:** $\mathcal{Z}_r = (1 - e^{-\beta(E_r-\mu)})^{-1}$

We can combine these into a single compact notation:

$$\ln \mathcal{Z} = \mp \sum_r \ln(1 \mp e^{-\beta(E_r-\mu)}) \tag{9.5}$$

where the upper sign $(-)$ refers to Bosons and the lower sign $(+)$ refers to Fermions.

9.2.2 Mean Occupation Numbers

The expected number of particles in a quantum state with energy E is given by:

$$\langle n(E) \rangle = \frac{1}{e^{\beta(E-\mu)} \pm 1} \tag{9.6}$$

where $(+)$ denotes Fermions and $(-)$ denotes Bosons.

9.3 Thermodynamics of Non-Interacting Quantum Gases

We now apply this to a 3D gas of non-interacting particles with spin S.

9.3.1 Density of States

To perform thermodynamic calculations, we must convert the sum over quantum states $\sum_k$ into an integral over energy $\int g(E)dE$.

For a particle in a box of volume $V = L^3$, the density of states in k-space (including spin degeneracy $2S+1$) is:

$$g(k)dk = \frac{(2S+1)Vk^2}{2\pi^2}dk \tag{9.7}$$

Using the dispersion relation $E = \frac{\hbar^2 k^2}{2m}$, we find $k = \sqrt{\frac{2mE}{\hbar^2}}$ and $dk \propto E^{-1/2}dE$. Substituting this back, we get the energy density of states:

$$g(E)dE = \frac{(2S+1)V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} E^{1/2}dE \tag{9.8}$$

9.3.2 The Grand Potential (Φ_G)

The Grand Potential is defined as $\Phi_G = -k_B T \ln \mathcal{Z}$. Converting the sum to an integral:

$$\Phi_G = \mp k_B T \int_0^\infty g(E) \ln(1 \pm e^{-\beta(E-\mu)})dE \tag{9.9}$$

Integrating by parts, we arrive at a fundamental relation for non-relativistic quantum gases:

$$\Phi_G = -\frac{2}{3}\int_0^\infty \frac{Eg(E)dE}{e^{\beta(E-\mu)} \pm 1} = -\frac{2}{3}U \tag{9.10}$$

Since $\Phi_G = -pV$, this implies the equation of state: $pV = \frac{2}{3}U$.

9.4 Particle Number and Internal Energy

We define the **fugacity** $z = e^{\beta\mu}$. The macroscopic properties can be expressed as integrals involving z:

$$N = \int_0^\infty \frac{g(E)dE}{z^{-1}e^{\beta E} \pm 1} = \frac{(2S+1)V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^\infty \frac{E^{1/2}dE}{z^{-1}e^{\beta E} \pm 1} \tag{9.11}$$

$$U = \int_0^\infty \frac{Eg(E)dE}{z^{-1}e^{\beta E} \pm 1} = \frac{(2S+1)V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\int_0^\infty \frac{E^{3/2}dE}{z^{-1}e^{\beta E} \pm 1} \tag{9.12}$$

These integrals are related to the **Polylogarithm functions** (or Fermi/Bose integrals) $Li_n(z)$.

$$\int_0^\infty \frac{E^{n-1}dE}{z^{-1}e^{\beta E} \pm 1} = (k_B T)^n \Gamma(n)[\mp Li_n(\mp z)] \tag{9.13}$$

$$N = \frac{(2S+1)V}{\lambda_{th}^3}[\mp Li_{3/2}(\mp z)] \tag{9.14}$$

$$U = \frac{3}{2}NK_BT\frac{Li_{5/2}(\mp z)}{Li_{3/2}(\mp z)} \tag{9.15}$$

9.4.1 The Classical (Ideal) Limit

In the limit of high temperature or low density ($n\lambda_{th}^3 \ll 1$), the fugacity $z \ll 1$. The quantum distributions approach the Maxwell-Boltzmann distribution:

$$\langle n \rangle \approx e^{-\beta(E-\mu)} \tag{9.16}$$

In this limit, we recover the classical ideal gas law:

$$U \approx \frac{3}{2}Nk_BT \quad \text{and} \quad \Phi_G \approx -Nk_BT \tag{9.17}$$

9.5 The Degenerate Fermi Gas ($T = 0$)

Consider a gas of Fermions at absolute zero ($T \to 0$, $\beta \to \infty$).

The occupation number function becomes a step function (Heaviside function):

$$\lim_{T\to 0} \frac{1}{e^{\beta(E-\mu)}+1} = \theta(\mu - E) \tag{9.18}$$

This means all states with energy $E < \mu(0)$ are filled, and all states above are empty. We define the **Fermi Energy** E_F as the chemical potential at zero temperature:

$$E_F \equiv \mu(T = 0) \tag{9.19}$$

9.5.1 Fermi Momentum and Energy

We calculate the total number of particles by filling states up to the Fermi wavevector k_F:

$$N = \int_0^{k_F} g(k)dk = \frac{(2S+1)V}{2\pi^2}\int_0^{k_F} k^2 dk = \frac{(2S+1)V}{6\pi^2}k_F^3 \tag{9.20}$$

Solving for k_F:

$$k_F = \left(\frac{6\pi^2 n}{2S+1}\right)^{1/3} \quad \text{where } n = N/V \tag{9.21}$$

The Fermi Energy is therefore:

$$E_F = \frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{6\pi^2 n}{2S+1}\right)^{2/3} \tag{9.22}$$

9.5.2 Ground State Energy

The total internal energy at $T = 0$ is:

$$U_0 = \int_0^{E_F} E g(E) dE \tag{9.23}$$

Evaluating this integral yields the average energy per particle for a degenerate Fermi gas:

$$\frac{U_0}{N} = \langle E \rangle = \frac{3}{5}E_F \tag{9.24}$$

This non-zero energy at absolute zero is a direct consequence of the Pauli Exclusion Principle (degeneracy pressure).

9.6 The Photon Gas (Blackbody Radiation)

We begin by considering a gas of photons. This serves as an important prelude to Bose-Einstein Condensation because photons are bosons, yet they behave differently from massive bosons due to their dispersion relation and non-conserved number.

9.6.1 Key Properties

- **Dispersion Relation:** For photons, energy is linear with momentum: $E = \hbar\omega = \hbar kc$.
- **Spin & Polarization:** Photons have spin $S = 1$. However, because they are massless, the magnetic quantum number m_s can only take values ± 1 (corresponding to two polarization states). The $m_s = 0$ state is forbidden.
- **Chemical Potential:** Photon number N is **not conserved**; photons are constantly absorbed and emitted by the cavity walls. Equilibrium requires the free energy to be minimized with respect to N, which leads to $\mu = 0$.

9.6.2 Density of States Derivation

To find the thermodynamics, we first determine the density of states $g(E)$. The number of states in a volume element of k-space is:

$$g(k)dk = 2 \times \frac{V}{(2\pi)^3} 4\pi k^2 dk = \frac{Vk^2}{\pi^2} dk \tag{9.25}$$

The factor of 2 accounts for the two polarization directions. Using the dispersion relation $k = E/(\hbar c)$, we transform this to energy space:

$$g(E)dE = \frac{V}{\pi^2} \frac{E^2}{(\hbar c)^3} dE \tag{9.26}$$

Note: For photons, $g(E) \propto E^2$, whereas for massive particles $g(E) \propto \sqrt{E}$.

9.6.3 Alternative way for Photon gas Internal Energy

The total internal energy U is calculated by integrating the energy over the Bose-Einstein distribution (with $z = e^{\beta\mu} = 1$ since $\mu = 0$):

$$U = \int_0^\infty \frac{Eg(E)dE}{e^{\beta E} - 1} = \frac{V}{\pi^2(\hbar c)^3} \int_0^\infty \frac{E^3 dE}{e^{\beta E} - 1} \tag{9.27}$$

We define a dimensionless variable $x = \beta E$, so $E = k_B T x$:

$$U = \frac{V(k_B T)^4}{\pi^2(\hbar c)^3} \underbrace{\int_0^\infty \frac{x^3 dx}{e^x - 1}}_{\Gamma(4)\zeta(4)} \tag{9.28}$$

Using the standard integral value $\Gamma(4)\zeta(4) = 6 \times \frac{\pi^4}{90} = \frac{\pi^4}{15}$, we recover the Stefan-Boltzmann law:

$$U = \frac{\pi^2 V(k_B T)^4}{15(\hbar c)^3} \implies U \propto T^4 \tag{9.29}$$

9.7 Bose-Einstein Condensation (BEC)

We now consider a gas of **massive** bosons (e.g., ^{4}He atoms) where N is conserved.

- Mass $m \neq 0$, so $E = \frac{\hbar^2 k^2}{2m}$.
- N is fixed, so $\mu \neq 0$.
- Stability requires $\mu \leq 0$ (otherwise occupation numbers would be negative).

9.7.1 The Conundrum: Breakdown of the Integral Approximation

Standard statistical mechanics tells us to calculate the total number of particles by integrating the density of states times the distribution function:

$$N = \int_0^\infty g(E)\langle n(E)\rangle dE = \int_0^\infty \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2} \frac{\sqrt{E}}{e^{\beta(E-\mu)}-1} dE \tag{9.30}$$

However, this mathematical approach hides a subtle but critical flaw.

Why the Math Breaks Down

As we lower the temperature T, the chemical potential μ must increase toward 0 to keep N constant. The maximum number of particles the excited states can hold occurs when we set $\mu = 0$ exactly.
Let's evaluate the integral at this limit ($\mu \to 0$):

$$N_{integral}^{max} = \int_0^\infty \frac{g(E)}{e^{\beta E}-1} dE \propto \int_0^\infty \frac{\sqrt{E}}{e^{\beta E}-1} dE \tag{9.31}$$

Notice the numerator $\sqrt{E}$. At the ground state ($E = 0$), the density of states $g(0) = 0$.
The Consequence: The integral assigns **zero weight** to the ground state. It effectively counts only the particles in the excited states ($E > 0$).
If we cool the gas below a critical temperature, the actual N exceeds the capacity of the excited states ($N > N_{integral}^{max}$). The "excess" particles must exist, but they disappear from our integral. Where are they?
Answer: They have "condensed" into the single quantum state at $E = 0$. To describe this, we must manually separate the ground state term from the integral.

9.7.2 The Two-Fluid Model

To resolve the conundrum, we write the total number of particles as the sum of the ground state population (N_0) and the excited state population (N_{ex}):

$$N = N_0 + N_{ex} \tag{9.32}$$

1. **Ground State (N_0):** This is the macroscopic occupation of the lowest level.

$$N_0 = \frac{1}{e^{-\beta\mu}-1} \tag{9.33}$$

2. **Excited States (N_{ex}):** This can be accurately calculated using the integral approximation.

$$N_{ex} = \frac{(2S+1)V}{\lambda_{th}^3} Li_{3/2}(z) \tag{9.34}$$

9.7.3 Critical Temperature (T_c)

The phase transition occurs when the excited states are saturated (i.e., when $\mu = 0$ or $z = 1$). The maximum density of excited particles is:

$$\frac{N_{ex}}{V} = \frac{2S+1}{\lambda_{th}^3}\zeta(3/2) \quad \text{where } \zeta(3/2) \approx 2.612 \tag{9.35}$$

We define the critical temperature T_c as the temperature where the total density $n = N/V$ equals this saturation density:

$$n = \frac{2S+1}{\lambda_{th}(T_c)^3} \times 2.612 \tag{9.36}$$

Solving for T_c (using $\lambda_{th} = h/\sqrt{2\pi m k_B T}$):

$$k_B T_c = \frac{2\pi\hbar^2}{m}\left(\frac{n}{2.612(2S+1)}\right)^{2/3} \tag{9.37}$$

9.7.4 Condensate Fraction

For temperatures $T < T_c$, the number of excited particles is determined by the thermal wavelength at that temperature (with $\mu = 0$):

$$N_{ex}(T) = N\left(\frac{T}{T_c}\right)^{3/2} \tag{9.38}$$

Since $N_0 = N - N_{ex}$, the fraction of particles in the condensate is:

$$\frac{N_0}{N} = 1 - \left(\frac{T}{T_c}\right)^{3/2} \tag{9.39}$$

9.7.5 Internal Energy $U(T)$ Below T_c

$T \leq T_c, z \approx 1$
The particles in the condensate have $E = 0$, so they contribute zero to the total energy. The internal energy comes entirely from the excited particles.

$$U = \int_0^\infty Eg(E)n(E)dE \propto T^{5/2} \tag{9.40}$$

Specifically, using the polylogarithm ratio relation:

$$U = \frac{3}{2}Nk_BT\frac{\zeta(5/2)}{\zeta(3/2)}\left(\frac{T}{T_c}\right)^{3/2} \tag{9.41}$$

Combining the T terms gives the characteristic dependence:

$$U \propto T^{5/2} \quad \Longrightarrow \quad C_V = \left(\frac{\partial U}{\partial T}\right)_V \propto T^{3/2} \tag{9.42}$$

9.8 Summary of Quantum Statistics

Symbol	**Formula**
N, U (Gen.)	$N = \frac{(2S+1)V}{\lambda_{th}^3}[\mp Li_{3/2}(\mp z)]$ $U = \frac{3}{2}Nk_BT\frac{Li_{5/2}(\mp z)}{Li_{3/2}(\mp z)}$
$\langle n \rangle$	$\frac{1}{e^{\beta(E-\mu)} \pm 1} \quad \left(\begin{smallmatrix}+\text{Fermions} \\ -\text{Bosons}\end{smallmatrix}\right)$
$E_F(T=0)$	$\frac{\hbar^2}{2m}(3\pi^2 n)^{2/3}$
$\langle E \rangle_{Fermi}$	$\frac{3}{5}E_F$
Li_n Identity	$\int_0^\infty \frac{E^{n-1}dE}{z^{-1}e^{\beta E} \pm 1} = (k_BT)^n\Gamma(n)[\mp Li_n(\mp z)]$
BEC Condition	$n\lambda_{th}^3 \geq \zeta(3/2)(2S+1)$
N_0/N	$1 - \left(\frac{T}{T_c}\right)^{3/2}$
$U_{BEC}(T < T_c)$	$U = \frac{3}{2}Nk_BT\frac{\zeta(5/2)}{\zeta(3/2)}\left(\frac{T}{T_c}\right)^{3/2}$